

On-Chain Prediction Market

Blockchain Technologies 2 | Final Project



<https://github.com/AltynayA/onchain-prediction-market>



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Baldauren

Role: Smart Contracts + Tests

Key deliverables:

- PredictionMarket, CPMM,
- YulHelpers
- Unit, fuzz, invariant tests

Altynay

Role: Governance + Deploy + CI

Key deliverables:

- Governor, Timelock contract
- Deploy script
- Base Sepolia

Zhaniya

Role: Frontend + Sub

Key deliverables:

- Governor, Timelock contract
- Deploy script
- Base Sepolia

Project Overview

Key Features of the System

This project leverages **binary outcome markets** integrated with Chainlink oracles, utilizing ERC-1155 tokens, a fee vault for liquidity providers, and **DAO governance** for complete lifecycle management.

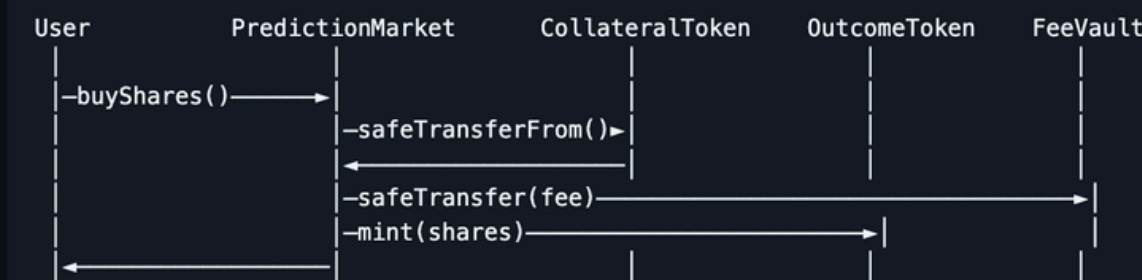
Architecture

- UUPS upgradeable PredictionMarket proxy
- ERC-1155 YES/NO outcome shares
- CPMM $x \cdot y = k$ AMM with 0.3% fee
- ERC-4626 FeeVault receives 1% of every trade
- Chainlink oracle with staleness check
- DAO controls all parameters via 2-day Timelock

State machine:

```
Active → Pending → Final
                ↘
                Disputed → Final
```

5.1 Buy Shares Flow



Market Factory

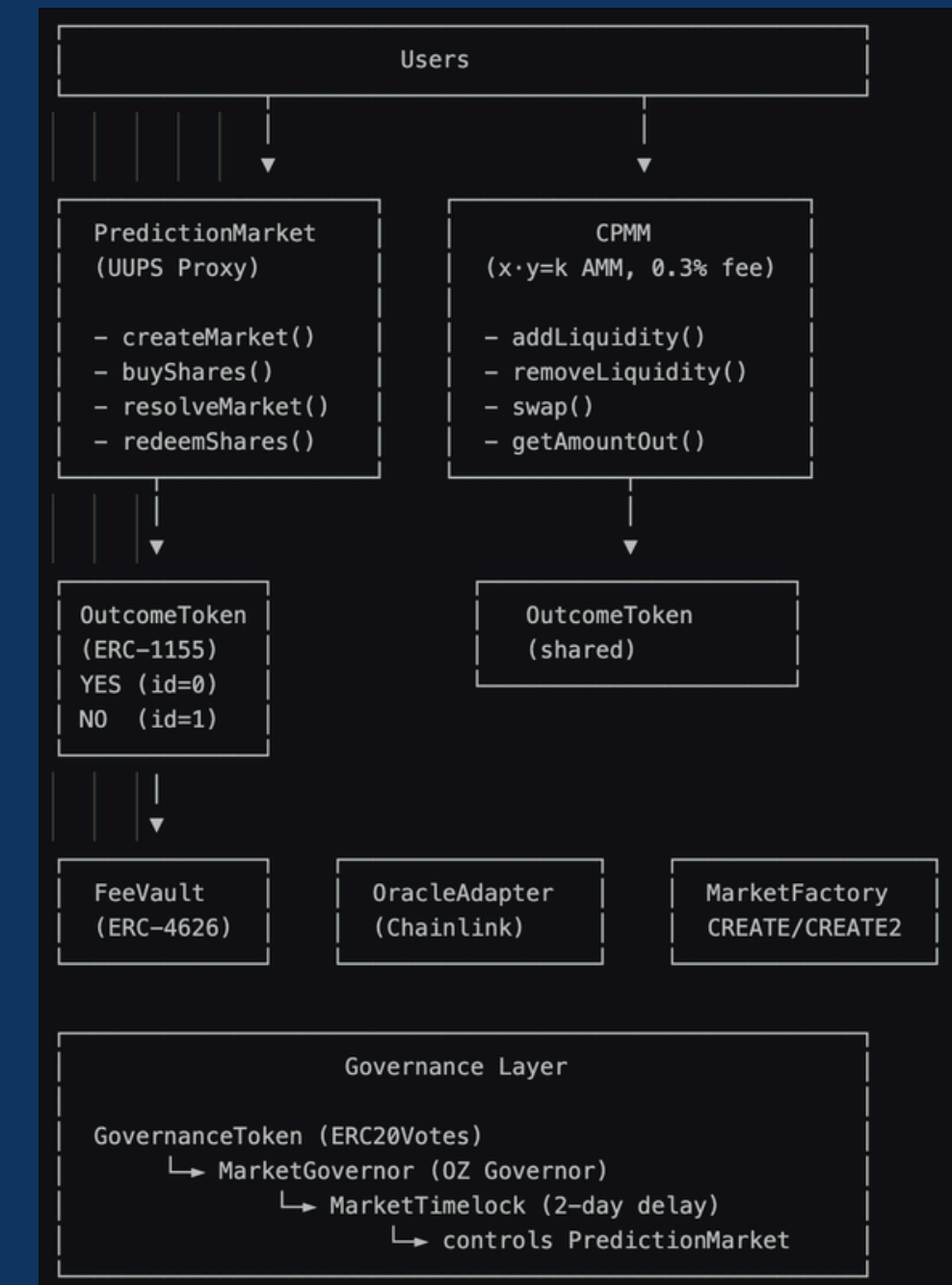
MarketFactory deploys isolated market instances via CREATE and CREATE2, ensuring deterministic addresses and independent liquidity pools per market.

Upgradeable Proxy

PredictionMarket uses UUPS proxy pattern - logic upgrades require DAO proposal + 2-day Timelock execution, making upgrades governance-controlled, not admin-controlled.

Token Management

Each market mints ERC-1155 YES (id=0) and NO (id=1) outcome shares. On resolution, winning shares redeem 1:1 for collateral. FeeVault accumulates 1% of every trade as ERC-4626 yield-bearing vault.



Smart Contracts + Design Patterns

10 design patterns:

- 1. UUPS
- 2. CEI
- 3. Factory
- 4. State Machine
- 5. AccessControl
- 6. Pausable
- 7. Oracle adapter
- 8. Timelock
- 9. ReentrancyGuard
- 10. Pull-over-push

Yul assembly in CPMM: 2-6% gas savings vs Solidity equivalent

```
contracts > core > CPMM.sol > ...
10  contract CPMM is ReentrancyGuard, IERC1155Receiver {
162      //internal math functions
163      function _getAmountOutYul(uint256 amountIn, uint256 resIn, uint256 resOut) internal pure returns (uint256 out) {
164          assembly {
165              let amountInWithFee := mul(amountIn, 997)
166              let numerator := mul(amountInWithFee, resOut)
167              let denominator := add(mul(resIn, 1000), amountInWithFee)
168              out := div(numerator, denominator)
169          }
170      }
```

MarketFactory supports CREATE and CREATE2

Contract 0xe81681ffc729181CbcB7943745DB79325136E8DE

Source Code

Overview

ETH BALANCE
0 ETH

More Info

CONTRACT CREATOR
0x90F79bf6...01E93b906 | 1 day ago

Multichain Info

N/A

Transactions

Internal Transactions

Token Transfers (ERC-20)

Contract

Events

Code

Read Contract

Write Contract

Heuristics suggest this may be a proxy contract. Verify using the Proxy Contract Checker

Source Code Verified

Exact Match

CONTRACT NAME	COMPILER VERSION	OPTIMIZATION ENABLED	OTHER SETTINGS	LICENSE
MarketFactory	v0.8.34+commit.80d5c536	Yes with 200 runs	prague EvmVersion	-NA-

Source

Similar Contracts

More Options

V1 -> V2 upgrade with no storage collision - storage gap documented

```
// storage gap so v2 and v3 releases don't collide
uint256[42] private __gapV2;
```

- ONCHAIN-PREDICTION-MARKET
 - .github / workflows
 - test.yml
 - broadcast
 - cache
 - contracts
 - core
 - CPMM.sol
 - FeeVault.sol
 - MarketFactory.sol
 - PredictionMarket.sol
 - PredictionMarketV2.sol
 - governance
 - MarketGovernor.sol
 - MarketTimelock.sol
 - PredictionToken.sol
 - mocks
 - oracles
 - OracleAdapter.sol
 - tokens
 - CollateralToken.sol
 - OutcomeToken.sol
 - utils
 - YulHelpers.sol



Testing

Coverage

test
fork
Fork.t.sol
fuzz
Fuzz.t.sol
integration
GovernanceE2E.t.sol
invariant
Invariant.t.sol
unit
Governance.t.sol
PredictionMarket.t.sol

Invariant tests

3 protocol guarantees that must hold across all states:

- k never decreases after swaps (CPMM)
- LP token supply matches tracked shares
- Total collateral always backs all outcome tokens

256 runs × 120 000 calls each

file	tests
PredictionMarket.t.sol	~61 unit
Fuzz.t.sol	10 fuzz (256 runs)
Invariant.t.sol	5 invariant
Fork.t.sol	4 fork

Fork tests — no mocks

Run against live Base Sepolia:

- Real Chainlink ETH/USD feed
- Real USDC contract on mainnet
- OracleAdapter reverts if stale > 3600s

Fuzz: AMM swap fee math Yul helpers



All passing in CI on every push

```
baldauren@MacBook-Air-Baldauren onchain-prediction-market % forge test --no-match-path "test/fork/*" -vvv
```

Contract	Selector	Calls	Reverts	Discards
CPMMHandler	swap	42586	0	0

Navigate to Command
Scroll to Previous Command (⌘↑)
Scroll to Next Command (⌘↓)

```
counts() (runs: 256, calls: 128000, reverts: 18243)
```

Contract	Selector	Calls	Reverts	Discards
CPMMHandler	addLiquidity	42729	0	0
CPMMHandler	removeLiquidity	42685	18243	0
CPMMHandler	swap	42586	0	0

```
[PASS] invariant_reserves_backed() (runs: 256, calls: 128000, reverts: 18358)
```

Contract	Selector	Calls	Reverts	Discards
CPMMHandler	addLiquidity	42729	0	0
CPMMHandler	removeLiquidity	42685	18358	0
CPMMHandler	swap	42586	0	0

```
Suite result: ok. 3 passed; 0 failed; 0 skipped; finished in 15.04s (43.35s CPU time)
```

```
Ran 6 tests for test/integration/GovernanceE2E.t.sol:GovernanceE2ETest
```

```
[PASS] test_directCall_toMarket_reverts_forNonTimeLock() (gas: 18256)
```

```
[PASS] test_governance_cannotExecuteBeforeTimeLockDelay() (gas: 300276)
```

```
[PASS] test_governance_defeated_againstWins() (gas: 200912)
```

```
[PASS] test_governance_defeated_quorumNotReached() (gas: 146658)
```

```
[PASS] test_governance_propose_vote_queue_execute() (gas: 388184)
```

```
[PASS] test_timelock_isAdminOfMarket() (gas: 16016)
```

```
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 15.04s (2.07ms CPU time)
```

```
Ran 2 tests for test/unit/Governance.t.sol:GovernanceTokenTest
```

```
[PASS] test_totalSupply_is1Million() (gas: 7566)
```

```
[PASS] test_votingPower_afterSelfDelegate() (gas: 86066)
```

```
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 20.40s (951.54µs CPU time)
```

```
Ran 2 tests for test/invariant/Invariant.t.sol:MarketInvariantTest
```

```
[PASS] invariant_collateral_backed() (runs: 256, calls: 128000, reverts: 0)
```

Contract	Selector	Calls	Reverts	Discards
MarketHandler	buyNo	64034	0	0
MarketHandler	buyYes	63966	0	0

```
[PASS] invariant_shares_conservation() (runs: 256, calls: 128000, reverts: 0)
```

Contract	Selector	Calls	Reverts	Discards
MarketHandler	buyNo	64034	0	0
MarketHandler	buyYes	63966	0	0

```
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 20.83s (41.21s CPU time)
```

```
Ran 8 test suites in 20.83s (71.48s CPU time): 76 tests passed, 0 failed, 0 skipped (76 total tests)
```

Security

PredictionMarket contract and test

Vulnerability 1: reentrancy

in redeemshares(), initially collateral transfer **before** share burns, so attacked can re-enter with onERC1155Received callback

Fix: CEI burn shares → update totalCollateral → transfer

```
function test_reentrancy_redeemShares_blocked() public {
    uint256 id = _buyAndResolve(true);
    _finalize(id);

    vm.prank(alice);
    outcomeToken.setApprovalForAll(address(market), true);

    uint256 aliceYes = outcomeToken.balanceOf(alice, 0);

    //first redeem succeeds
    vm.prank(alice);
    market.redeemShares(id, aliceYes);

    // second redeem reverts as shares already burned
    vm.prank(alice);
    vm.expectRevert();
    market.redeemShares(id, 1);
}
```

```
//effects
m.totalCollateral -= collateralOut;

// effects before interactions (CEI)
outcomeToken.burn(msg.sender, winningId, amount);

// interactions
collateral.safeTransfer(msg.sender, collateralOut);
emit SharesRedeemed(marketId, msg.sender, amount, collateralOut);
}
```

Vulnerability 2: access control

anyone could call resolveMarket()

```
// before fix, anyone could call resolveMarket
function test_resolveMarket_revert_notResolver() public {
    uint256 id = _createMarket();
    vm.warp(block.timestamp + RES_OFFSET + 1);
    vm.prank(attacker);
    vm.expectRevert();
    market.resolveMarket(id, true);
}
```

Fix: onlyRole(RESOLVER_ROLE) modifier

```
function resolveMarket(uint256 marketId, bool outcome) external onlyRole(RESOLVER_ROLE) nonReentrant whenNotPaused {
    Market storage m = markets[marketId];
    require(m.state == MarketState.Active, "PM: not active");
    require(block.timestamp >= m.resolutionTime, "PM: too early");

    IOracleAdapter(m.oracleAdapter).assertFresh(defaultStaleness);

    uint256 deadline = block.timestamp + defaultDisputeWindow;
    m.state = MarketState.Pending;
    m.outcome = outcome;
    m.disputeDeadline = deadline;

    emit MarketResolved(marketId, outcome, deadline);
}
```

Slither static analysis: 0 High, 0 Medium findings



```
Run Slither

1 ▶Run python3 -m slither contracts/ --config-file .slither.json --foundry-compile-all
7
7 'forge clean' running (wd: /home/runner/work/onchain-prediction-market/onchain-prediction-market)
8 'forge config --json' running
9 Warning: Selected profile `ci` does not exist; falling back to the default profile.
10 'forge build --build-info contracts' running (wd: /home/runner/work/onchain-prediction-market/onchain-prediction-market)
11 INFO:Slither:contracts/ analyzed (94 contracts with 57 detectors), 0 result(s) found
```

```
Ran 2 tests for test/unit/PredictionMarket.t.sol:PredictionMarketTest
[PASS] test_reentrancy_redeemShares_blocked() (gas: 405659)
[PASS] test_resolveMarket_revert_notResolver() (gas: 121654)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 5.29ms (1.38ms CPU time)
```

```
~/onchain-prediciton-market/ [main] forge test --match-path test/fork/Fork.t.sol \
--fork-url $MAINNET_RPC_URL -vv
[.] Compiling...
No files changed, compilation skipped
```

```
Ran 4 tests for test/fork/Fork.t.sol:ForkTest
[PASS] test_fork_feeVault_accumulatesFees_withRealUSDC() (gas: 1030704)
[PASS] test_fork_feeVault_depositAndRedeem_withRealUSDC() (gas: 1093844)
[PASS] test_fork_oracleAdapter_realChainlinkFeed() (gas: 246719)
[PASS] test_fork_predictionMarket_buyShares_withRealUSDC() (gas: 5285545)
Suite result: ok. 4 passed; 0 failed; 0 skipped; finished in 2.59s (4.62s CPU time)
```


Governance & Deployment

Governor stack:

- Voting delay: 1 day
- Voting period: 1 week
- Quorum: 4%
- Proposal threshold: 1%
- MarketTimelock: 2-day mandatory delay before execution

full governance lifecycle

```
~/onchain-prediciton-market/ [main] forge test --match-contract GovernanceE2ETest -vv
[.] Compiling...
No files changed, compilation skipped

Ran 6 tests for test/integration/GovernanceE2E.t.sol:GovernanceE2ETest
[PASS] test_directCall_toMarket_reverts_forNonTimelock() (gas: 18256)
[PASS] test_governance_cannotExecuteBeforeTimelockDelay() (gas: 300276)
[PASS] test_governance_defeated_againstWins() (gas: 200912)
[PASS] test_governance_defeated_quorumNotReached() (gas: 146658)
[PASS] test_governance_propose_vote_queue_execute() (gas: 388184)
[PASS] test_timelock_isAdminOfMarket() (gas: 16016)
Suite result: ok. 6 passed; 0 failed; 0 skipped; finished in 4.67ms (1.90ms CPU time)

Ran 1 test suite in 106.29ms (4.67ms CPU time): 6 tests passed, 0 failed, 0 skipped (6 total tests)
```

```
docs > verify-output.txt
82 Script ran successfully.
83
84 == Logs ==
85 post-deployment verification
86 timelock is market admin: true
87 deployer has market admin: false
88 timelock min delay (seconds): 172800
89 governor is timelock proposer: true
90 deployer has timelock admin: false
91 voting delay (seconds): 86400
92 voting period (seconds): 604800
93 quorum fraction (%): 4
94 oracle staleness (seconds): 3600
95 dispute window (seconds): 172800
96 all checks passed
97
```



Verify.s.sol
post-deploymentcheck
Base Sepolia (chain 84532)



PredictionMarket (proxy)

Contract 0x0cEC8fd1D75bFe985807BeEEc1ACdaBE1784bB7

Source Code (Proxy)

Implementation: 0x712F373B...7F25F6d9C

Overview

ETH BALANCE
0 ETH

More Info

CONTRACT CREATOR
0x90F79bf6...01E93b906 | 1 day ago ↗

Multichain Info

N/A

Transactions

Token Transfers (ERC-20)

Contract

Events

Latest 2 from a total of 2 transactions

Download Page Data

Transaction Hash	Method	Block	Age	From	To	Amount	Txn Fee
0x6ad48fe741a...	Revoke Role	41580695	33 hrs ago	0x90F79bf6...01E93b906	0x0cEC8fd1...BE1784bB7	0 ETH	0.00000019
0x7c237d5e6d...	Grant Role	41580694	33 hrs ago	0x90F79bf6...01E93b906	0x0cEC8fd1...BE1784bB7	0 ETH	0.00000033

MarketTimelock

Contract 0xE699943a7d8b351E8c30064dB3511278D5f58042

Source Code

Overview

ETH BALANCE
0 ETH

More Info

CONTRACT CREATOR
0x90F79bf6...01E93b906 | 1 day ago ↗

Multichain Info

N/A

Transactions

Token Transfers (ERC-20)

Contract

Events

Latest 3 from a total of 3 transactions

Download Page Data

Transaction Hash	Method	Block	Age	From	To	Amount	Txn Fee
0x8483916548f...	Renounce Role	41580696	35 hrs ago	0x90F79bf6...01E93b906	0xE699943a...8D5f58042	0 ETH	0.00000014
0xe0b4a40d57...	Grant Role	41580691	35 hrs ago	0x90F79bf6...01E93b906	0xE699943a...8D5f58042	0 ETH	0.00000003
0x6a35d9e70c...	Grant Role	41580690	35 hrs ago	0x90F79bf6...01E93b906	0xE699943a...8D5f58042	0 ETH	0.00000003

Deployed Contracts — Base Sepolia (chain 84532)

Contract	Address
CollateralToken (mUSDC)	0x8fAC0f9B73DaC33f6d981A58a2De2C9a61A5d50C
OutcomeToken (ERC-1155)	0x8A6A50Db75992c17e9adFde99818921757521B3e
FeeVault (ERC-4626)	0x89a375715161F7221a890eB7dE11086d9Aa0DE84
PredictionMarket (proxy)	0x0cEC8fd1D75bFe985807BeEEc1ACdaBE1784bB7
PredictionMarket (impl)	0x712F373B113ea2C1d8880aF01d388DA7F25F6d9C
MarketFactory	0xe81681ffC729181Cbcb7943745DB79325136E8DE
CPMM	0x903FB4d0A0c7FA3a076D900039737B0b059A1d79
GovernanceToken (PRED)	0xA7086085727494305CD37E13adF446c096B24e70
MarketTimelock	0xE699943a7d8b351E8c30064dB3511278D5f58042
MarketGovernor	0xBdF66e0A87a5759444ec0b5858BDf581Ce3Ef309E

All contracts verified on [Basescan](#).

Frontend

- Plain HTML/JS with Ethers.js — no framework, no build step
- MetaMask wallet connection + automatic Base Sepolia network detection
- Dashboard reads on-chain: mUSDC balance, GOV balance, voting power, vault shares, pool reserves
- Market page: buy YES/NO outcome shares (Write TX #1)
- Vault page: deposit collateral to ERC-4626 FeeVault (Write TX #2)
- Governance page: delegate votes (Write TX #3)
- All 3 write transactions confirmed on Base Sepolia

Prediction Market

Dashboard Market Vault Governance 0x90F7...b906

Dashboard

Wallet

Address: 0x90F7...b906

mUSDC Balance: 1998.00 mUSDC

GOV Balance: 1000000.00 GOV

Voting Power: 1000000.00 GOV

Delegating to: 0x90F7...b906

AMM Pool Reserves

YES Reserve: 0.0000 shares

NO Reserve: 0.0000 shares

YES Probability: 0.00%

Vault Total Assets: 2.00 mUSDC

Your Vault Shares: 2.0000 vPRED

Prediction Market Dashboard Market Vault Governance 0x90F7...b906

Fee Vault (Write TX #2)

Vault Info

Total Assets: 2.00 mUSDC

Your Shares: 2.0000 vPRED

Your mUSDC Balance: 1998.00 mUSDC

Refresh

DAO Governance

Your Voting Power

Balance: 1000000.00 GOV

Voting Power: 1000000.00 GOV

Delegating to: 0x90F7...b906

Delegate Votes (Write TX #3)

0x90F79bf6EB2c4f870365E785982E1f101E93b906

Delegate

Delegated! TX: 0xf1076126...

Одобрить лимит расходов в размере ...
Подтверждено

Mint -0 ETH
Подтверждено -0,00 ETH

May 17, 2026

Delegate -0 ETH
Подтверждено -0,00 ETH

Deposit -0 ETH
Подтверждено -0,00 ETH

Одобрить лимит расходов в размере ...
Подтверждено

Одобрить лимит расходов в размере ...
Подтверждено

Mint -0 ETH
Подтверждено -0,00 ETH



Subgrapgh+docs

- Subgraph deployed on Goldsky — Base Sepolia, status: Live
- Indexes: MarketCreated · SharesBought · MarketResolved · MarketDisputed · MarketFinalized
- Governance page reads proposals via GraphQL (not direct contract calls)
- Architecture doc: 6 pages, C4 diagrams, 3 sequence diagrams, storage layout, 5 ADRs
- Audit report: 8 pages, VULN-01 + VULN-02 reproduced and fixed, Slither clean
- Gas report: Yul assembly 2-6% savings vs Solidity, L1 vs L2 comparison

marketCreated

block:

first: 0

orderBy:

orderDirection:

skip:

subgraphError:

where:

block_number

contractId_

id

marketId

oracleAdapter

question

resolutionTime

timestamp_

transactionHash_

sharesBought

block:

first: \$ 10

orderBy: \$ id

orderDirection:

skip:

subgraphError:

where:

block_number

buyer

collateralIn

contractId_

id

isYes

marketId

sharesMinted

timestamp_

transactionHash_

..

architecture.md

audit-report.md

coverage-report.md

gas-report.md

verify-output.md

verify-output.txt

Status

• Live

Block 41580683

Block 41644387 / 41644387